

MUSKOGEE-DAVIS, OK, USA

WMO: 723556

Lat: 35.657N

Lon: 95.361W

Elev: 186

StdP: 99.11

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 93953

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.1	-15.2	1.0	-6.8	-12.6	1.3	-4.4	11.9	8.5	11.0	8.6	3.9	350	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	37.6	24.3	35.9	24.5	33.9	24.5	26.9	32.3	26.2	32.1	25.6	31.7	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	21.1	29.9	24.5	19.9	29.2	23.9	19.2	28.6	85.0	32.0	82.1	32.1	79.6	32.0	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.3	8.4	-14.1	39.1	3.1	2.6	-16.3	41.0	-18.1	42.6	-19.8	44.0	-22.0	45.9		
(5)			-14.1	27.7	3.3	1.2	-16.5	28.5	-18.5	29.2	-20.3	29.9	-22.7	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	4.0	6.2	11.4	16.0	20.5	25.0	27.5	27.4	23.4	16.8	10.7	5.1		
	DBStd	9.48	5.78	6.06	5.40	4.53	3.81	2.65	2.61	2.80	3.72	4.88	5.24	5.36		
	HDD10.0	619	199	135	47	7	1	0	0	0	0	6	54	169		
	HDD18.3	1863	444	341	222	96	23	1	0	0	5	88	232	411		
	CDD10.0	2893	14	28	91	187	327	452	543	540	402	217	76	17		
	CDD18.3	1094	0	1	7	26	91	202	284	282	157	41	3	1		
	CDH23.3	11349	1	8	59	219	741	1991	3261	3165	1504	365	33	1		
	CDH26.7	5252	0	1	8	41	203	851	1686	1675	672	111	3	0		
(14) Wind	WSAvg	3.9	4.3	4.5	4.7	4.7	4.0	3.6	3.0	3.0	3.2	3.7	4.2	4.1		
(15) Precipitation	PrecAvg	1106	59	52	95	117	143	125	87	70	122	88	88	70		
	PrecMax	1503	153	180	238	224	308	305	324	162	295	271	201	176		
	PrecMin	691	6	0	21	42	35	32	4	0	16	16	6	8		
	PrecStd	189	38	39	52	52	64	70	69	39	68	62	66	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.1	27.5	30.0	32.4	36.1	39.7	40.8	37.9	32.3	27.0	22.4		
		MCWB	14.2	15.8	17.5	19.3	23.7	23.6	23.7	24.0	23.8	22.3	18.1	17.3		
	2%	DB	19.1	21.5	25.0	27.6	30.5	33.7	37.4	37.8	34.1	29.8	23.8	19.2		
		MCWB	13.2	14.8	17.0	19.4	23.0	24.6	24.4	24.4	23.4	21.2	16.6	14.5		
	5%	DB	16.9	18.8	22.7	26.0	28.9	32.5	35.5	36.2	32.6	27.4	21.9	16.8		
		MCWB	11.8	13.7	15.5	18.1	22.2	24.4	24.7	24.6	23.3	19.3	16.2	12.2		
	10%	DB	13.8	16.4	20.6	24.0	27.4	31.3	33.7	34.0	30.9	25.0	19.4	13.8		
		MCWB	8.9	11.5	14.6	17.4	21.6	24.2	25.1	24.4	22.8	18.6	14.3	9.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	18.4	19.6	22.4	25.3	27.4	28.0	27.3	26.1	23.9	20.3	18.4		
		MCDB	19.6	22.4	25.4	26.9	30.1	32.1	32.9	32.7	32.9	29.2	24.1	21.5		
	2%	WB	15.0	16.7	18.3	20.8	24.1	26.2	27.1	26.4	25.1	22.7	18.5	15.9		
		MCDB	17.9	19.7	23.4	25.4	28.7	31.6	32.3	32.8	31.8	27.5	22.0	18.5		
	5%	WB	12.1	14.6	17.0	19.9	23.3	25.5	26.4	25.7	24.3	21.5	17.1	13.1		
		MCDB	15.8	18.1	21.2	23.9	27.6	30.7	32.0	32.5	30.2	25.0	20.4	15.3		
	10%	WB	9.3	11.9	15.5	18.7	22.3	24.7	25.8	25.2	23.6	20.1	15.4	10.4		
		MCDB	13.4	16.2	19.7	22.7	26.3	29.7	31.5	32.2	28.8	23.6	18.8	13.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	11.4	11.9	11.9	10.8	10.5	11.2	11.9	12.1	12.5	12.1	10.9		
		MCDBR	16.1	14.9	14.7	14.2	12.0	12.0	13.7	14.7	13.9	14.4	14.6	15.1		
	5% WB	MCWBR	10.2	9.7	8.3	7.6	5.9	4.8	3.9	4.3	5.0	6.8	8.9	9.9		
		MCWBR	13.2	13.4	12.2	11.1	10.4	10.5	10.9	12.0	11.7	11.1	12.5	11.7		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.314	0.354	0.389	0.425	0.429	0.447	0.448	0.396	0.352	0.322	0.301		
	taud		2.485	2.483	2.401	2.315	2.275	2.292	2.287	2.255	2.364	2.472	2.505	2.515		
	Ebn at Noon		896	931	916	895	863	856	838	830	860	874	871	881		
	Edh at Noon		83	94	111	127	135	132	132	134	114	94	81	76		
(44) All-Sky Solar Radiation	RadAvg		2.56	3.08	4.11	5.11	5.63	6.44	6.53	5.93	4.96	3.72	2.73	2.17		
	RadStd		0.26	0.45	0.26	0.31	0.51	0.68	0.40	0.38	0.40	0.51	0.33	0.28		

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page